

Mobile Computing

Unit I: Introduction to Mobile Computing

Duration: 5 Hours

Topics

- What is Computing?
- What is Mobility?
- What is Mobile Computing?
- Definitions of Mobile Computing
- Mobile Computing Structure
 - Mobile Communication
 - Mobile Hardware
 - Mobile Software

Learning Outcomes

Students will be able to:

- Define mobile computing.
 - Explain components of mobile computing.
 - Describe advantages and applications of mobile computing.
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Unit II: Mobile Devices and Operating Systems

Duration: 8 Hours

Topics

- Mobile Hardware
- Mobile Devices
- Symbian OS
- Android OS
- Blackberry OS
- Windows Phone
- Palm OS
- Mobile Linux

Learning Outcomes

Students will be able to:

- Compare different mobile operating systems.
 - Explain features of Android and other mobile platforms.
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Unit III: Mobility and Communication

Duration: 8 Hours

Topics

- User Mobility
- Device Portability
- Wireless Communication
- Mobile Network Infrastructure
- Mobile Services

Learning Outcomes

Students will be able to:

- Explain mobility concepts.
 - Describe wireless communication mechanisms.
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Unit IV: Mobile Application Development

Duration: 10 Hours

Topics

- Mobile Application Development Lifecycle
- Requirement Analysis
- Platform Selection
- Prototype Development
- Application Integration
- Deployment

Learning Outcomes

Students will be able to:

- Develop mobile application plans.
 - Design and prototype simple mobile applications.
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Unit V: Mobile Testing and Advanced Applications

Duration: 10 Hours

Topics

- Mobile Application Testing
- Device Compatibility Testing
- Mobile Security
- Text-to-Speech (TTS)
- Speech Synthesis
- Emerging Mobile Technologies

Learning Outcomes

Students will be able to:

- Perform mobile application testing.
 - Explain speech technologies used in mobile systems.
 - Discuss future trends in mobile computing.
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